Final Exam

PhD Macro

Instructions. Show steps. Partial credit for correct intermediate steps.

Question 1: human capital investment using goods [35 pts]

In the "human capital investment using goods" model (Topic 4, Case 4), the planner chooses next-period stocks (K', X') and consumption C subject to

$$C = F(K, X) + (1 - \delta)K + (1 - \delta_x)X - K' - X',$$

with $\delta \in (0, 1)$ and $\delta_x \in (0, 1)$. Assume Cobb–Douglas technology

$$F(K, X) = K^\alpha X^{1-\alpha}, \qquad \alpha \in (0, 1),$$

and CRRA utility

$$u(C) = \frac{C^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma > 0.$$

Let $k \equiv K/X$ and write $F(K, X) = Xf(k)$ where $f(k) = k^\alpha$.

(a) **BGP condition for k^*.** [14 pts]
Using the slide result that an interior BGP with $K' > 0$ and $X' > 0$ implies

$$f'(k_{t+1}) - \delta = f(k_{t+1}) - k_{t+1}f'(k_{t+1}) - \delta_x,$$

and imposing $k_{t+1} = k_t = k^*$ on the BGP, show that k^* solves

$$(1 + k)f'(k) - f(k) = \delta - \delta_x. \tag{3}$$

For Cobb–Douglas $f(k) = k^\alpha$, show that when $\delta = \delta_x$ the unique solution is

$$k^* = \frac{\alpha}{1 - \alpha}.$$

(If $\delta \neq \delta_x$, k^* is obtained by solving (3) numerically, as done in the slides.)

(b) **Growth rate of growing series.** [10 pts]
Using the slide formula for the growth rate of growing series,

$$\gamma_Y = \cdots = \gamma_C = \left[\beta\left(f'(k^*) + 1 - \delta\right)\right]^{1/\sigma},$$

compute γ_Y for

$$\alpha = 0.30, \quad \delta = 0.10, \quad \delta_x = 0.15, \quad \beta = 0.985, \quad \sigma = 2,$$

using the slide-reported solution $k^* = 0.42857$.

(c) **Interpretation (one line).** [11 pts]
Holding $(\alpha, \beta, \sigma, \delta)$ fixed, what is the direction of change in k^* when δ_x increases? Briefly explain using equation (3).

Question 2: Lucas (1988) human capital accumulation model **[30 pts]**

In the Lucas (1988) model (Topic 4), specialize production to

$$Y_t = K_t^{\alpha} M_t^{1-\alpha}, \qquad \alpha \in (0,1),$$

where $M_t = \phi_t X_t$, X_t is the stock of skills, and $\phi_t \in (0,1)$ is the fraction of unit time endowment spent working. The fraction $1-\phi_t$ is time in education. On a balanced growth path with constant ϕ, skills grow according to the slide equation:

$$\gamma_X \equiv \frac{X_{t+1}}{X_t} = 1 + A(1-\phi), \qquad A > 0.$$

Preferences are CRRA with parameter σ and discount factor β.

(a) **BGP growth rate.** [12 pts]
Using the slide result, state and use the key implication:

$$\gamma_Y = \cdots = \gamma_X = \left[\beta(1+A)\right]^{1/\sigma}.$$

(b) **Solve for ϕ.** [10 pts]
Combine $\gamma_X = 1 + A(1-\phi)$ with the BGP growth rate to show:

$$\phi = \frac{1}{A}\left[1 + A - \left(\beta(1+A)\right)^{1/\sigma}\right].$$

(c) **Numerical.** [8 pts]
Compute ϕ, γ_X, and γ_Y when

$$\beta = 0.96, \quad \sigma = 2, \quad A = 0.20.$$

Question 3: aggregation over heterogenous firms and $A(\mu)$ **[35 pts]**

There is a large number of heterogeneous firms. Production at each firm is

$$y = z\varepsilon k^{\alpha} n^{\nu}, \tag{1}$$

where $\alpha \in (0,1)$, $\nu \in (0,1)$, and $\alpha + \nu < 1$. Capital is rented and labor is hired. Profits are

$$\pi(z\varepsilon; r, w) \equiv \max_{k,n} \; z\varepsilon k^{\alpha} n^{\nu} - (r+\delta)k - wn, \tag{2}$$

where r is the real interest rate, w is the real wage, and $\delta \in (0,1)$. Optimal choices satisfy

$$\alpha z\varepsilon k^{\alpha-1} n^{\nu} = r + \delta, \qquad \nu z\varepsilon k^{\alpha} n^{\nu-1} = w.$$

Let $\mu(\varepsilon)$ describe the measure of firms in production over ε.

(a) **Firm choices (equations (6)–(8)).** [12 pts]
Dividing (3) by (4) gives equation (6):

$$\frac{\alpha}{\nu}\frac{n}{k} = \frac{r+\delta}{w}. \tag{6}$$

Using (6) in (3), show that capital demand is (equation (7) in the slides):

$$k(\varepsilon) = \left(\frac{z\varepsilon\, \alpha^{1-\nu}\nu^{\nu}}{(r+\delta)^{1-\nu} w^{\nu}} \right)^{\frac{1}{1-(\alpha+\nu)}}. \tag{7}$$

(You do not need to re-derive the labor demand formula (8).)

(b) **Market clearing and the key moment.** [12 pts]
Using the slide market-clearing equations (10)–(11),

$$K = \kappa_K(z, r+\delta, w) \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}}\, \mu(d\varepsilon), \tag{10}$$

$$N = \kappa_N(z, r+\delta, w) \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}}\, \mu(d\varepsilon), \tag{11}$$

define

$$\mathcal{M}(\mu) \equiv \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}}\, \mu(d\varepsilon).$$

Explain why, holding $(z, r+\delta, w)$ fixed, changes in μ affect K and N only through $\mathcal{M}(\mu)$.

(c) **Distribution changes and aggregate productivity.** [11 pts]
Define an "endogenous TFP" index

$$A(\mu) \equiv \left(\mathcal{M}(\mu)\right)^{1-(\alpha+\nu)}.$$

Let $\alpha = 0.30$, $\nu = 0.50$ so $1-(\alpha+\nu) = 0.20$. Compare:

$$\mu_1 :\ \varepsilon = 1 \text{ w.p. } 1, \qquad \mu_2 :\ \varepsilon = 0.5 \text{ w.p. } 0.5,\ \varepsilon = 1.5 \text{ w.p. } 0.5.$$

Compute $A(\mu_1)$ and $A(\mu_2)$. Does greater dispersion raise or lower $A(\mu)$ here?